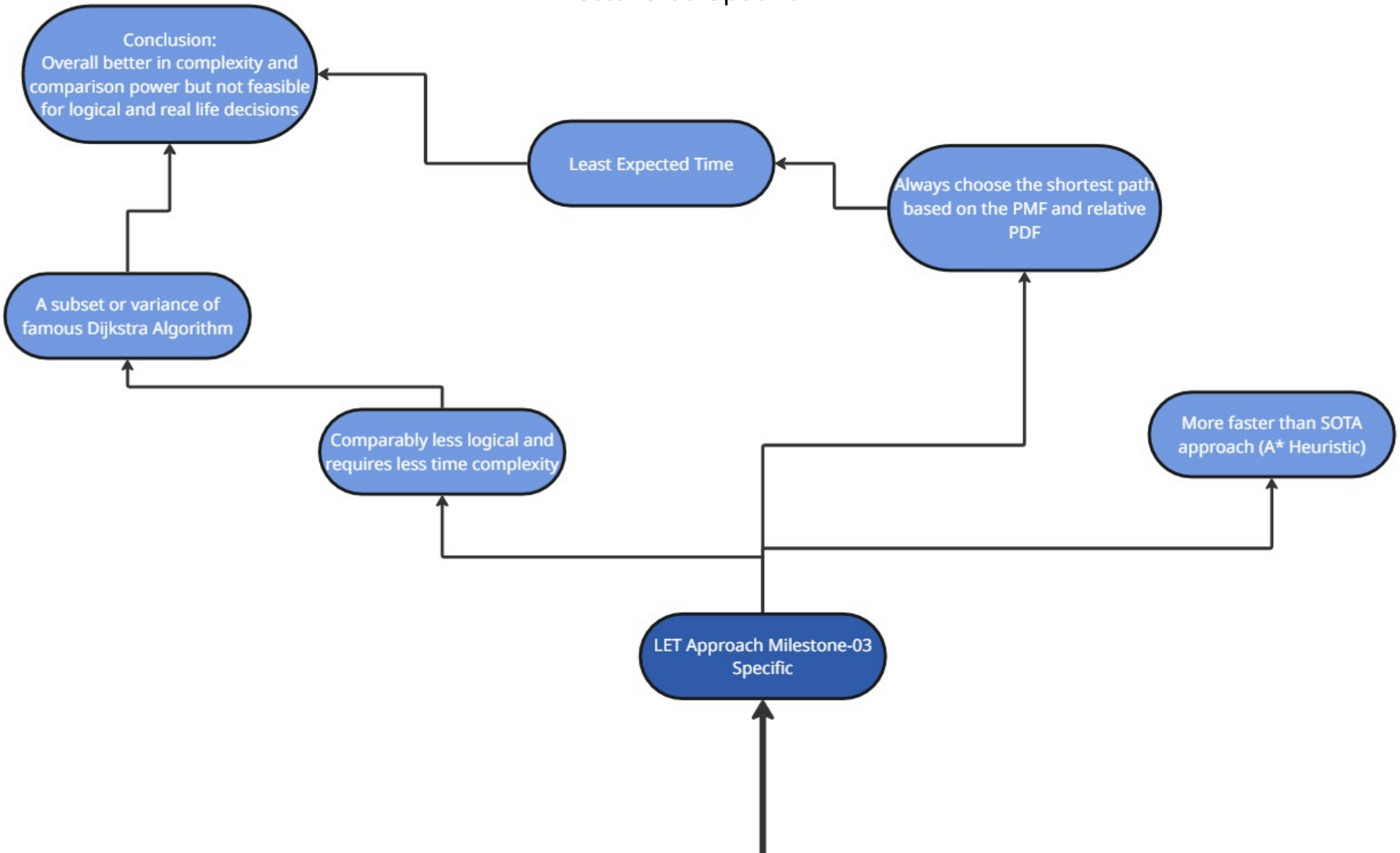
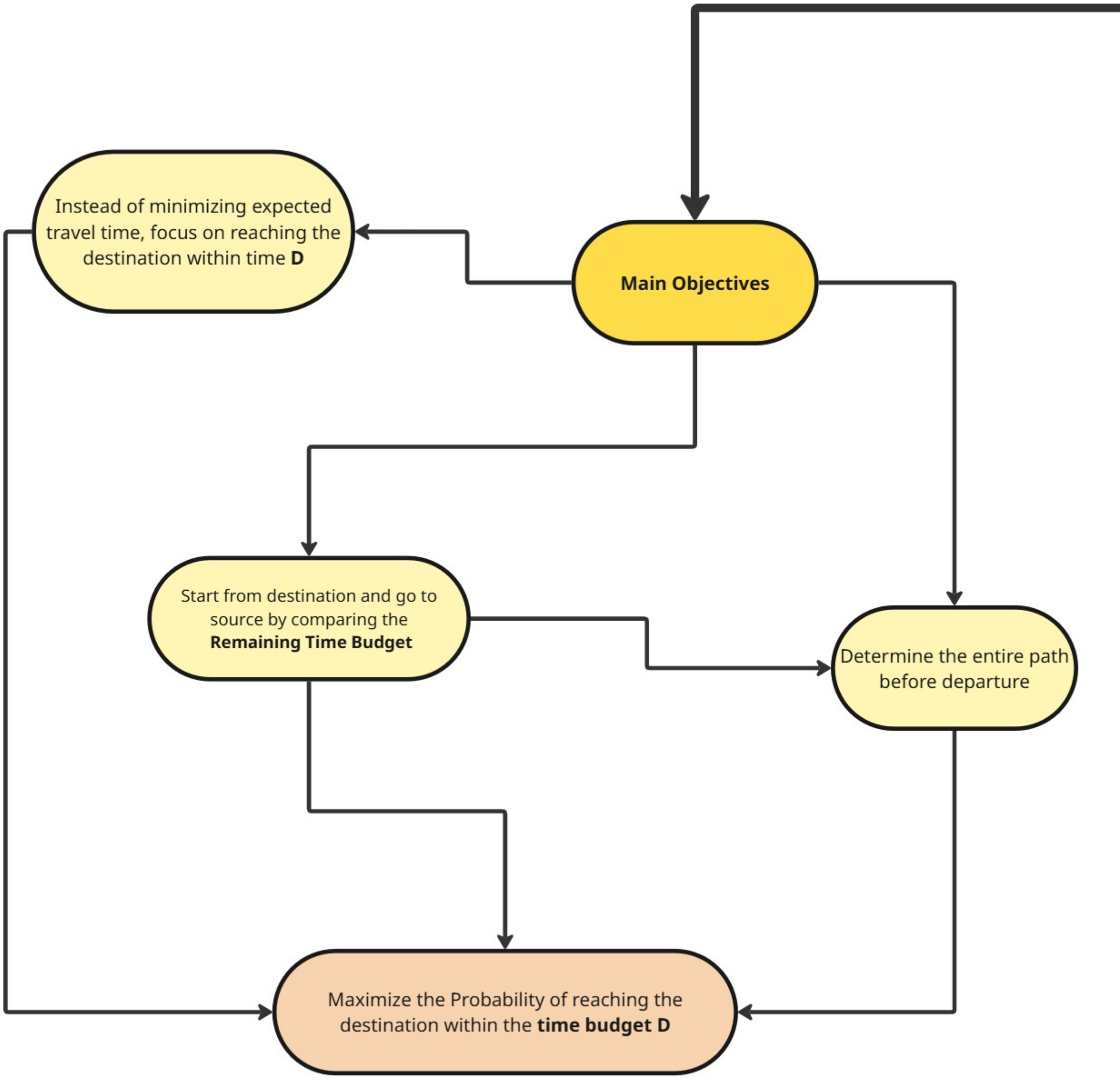
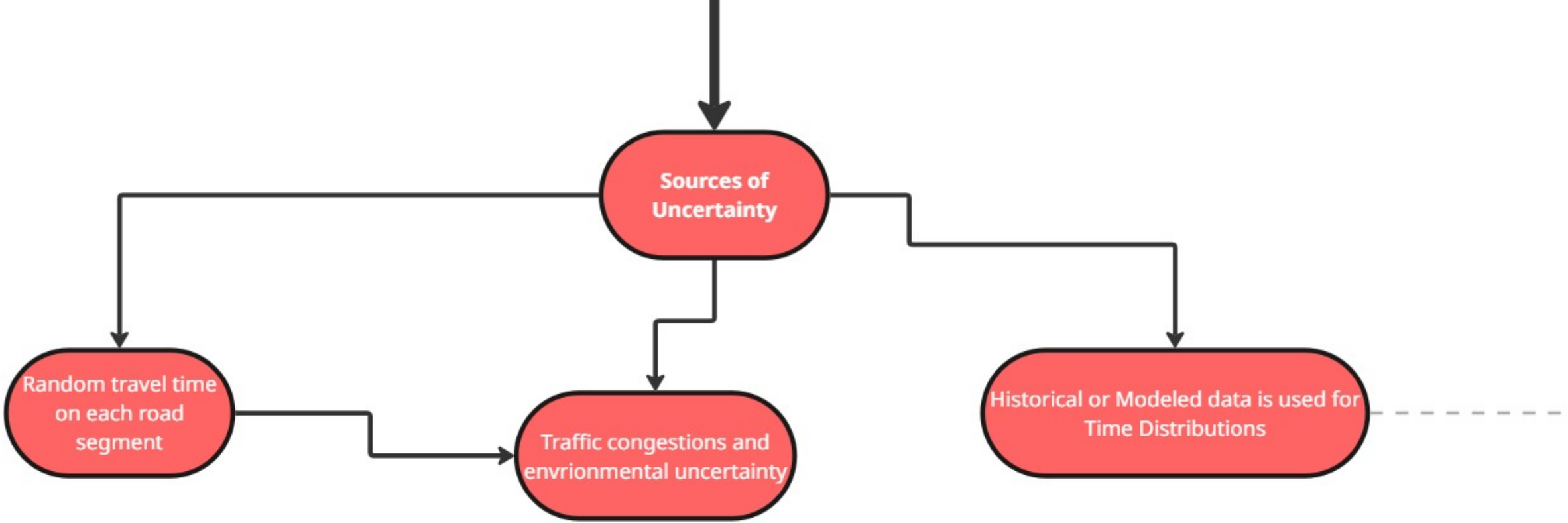
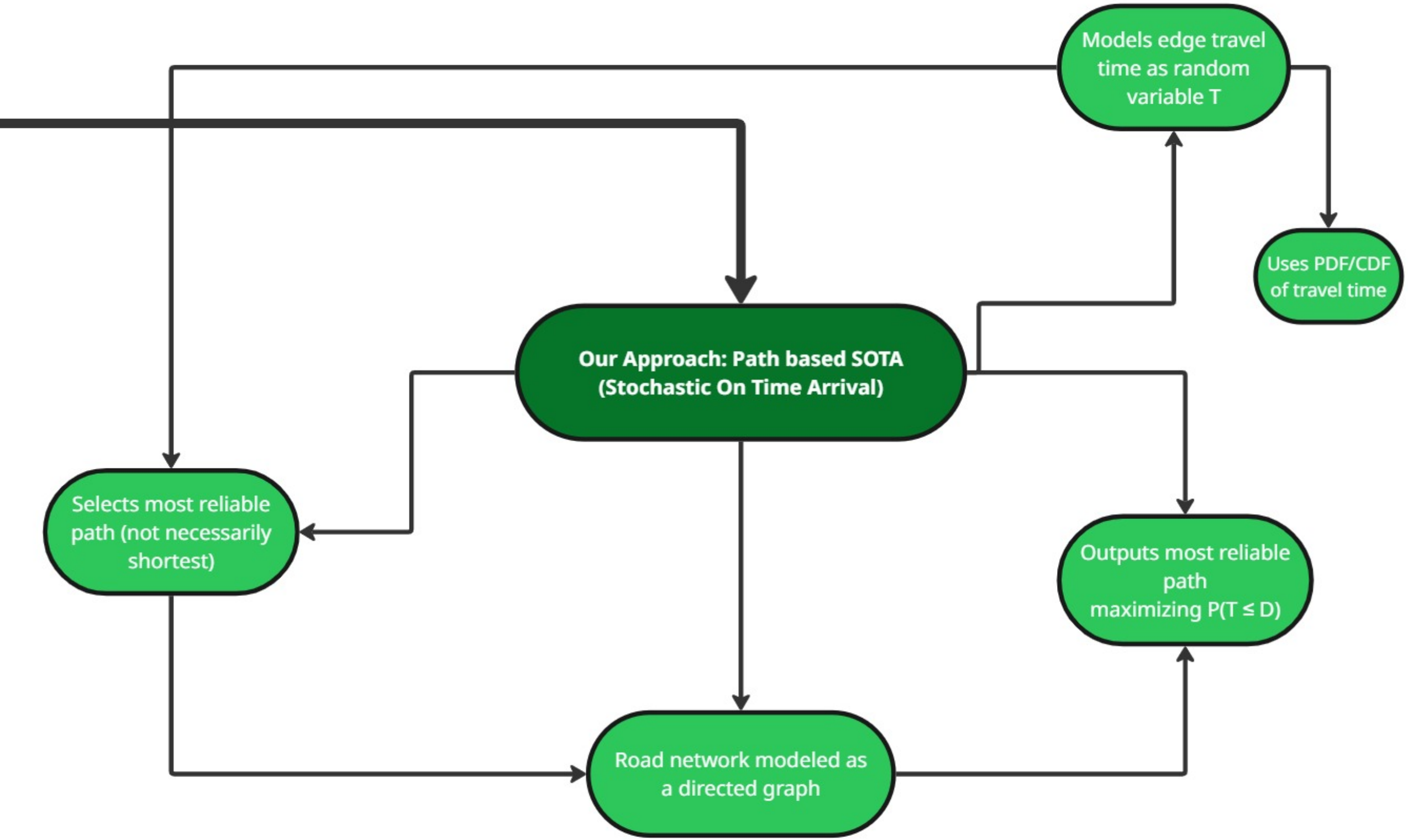
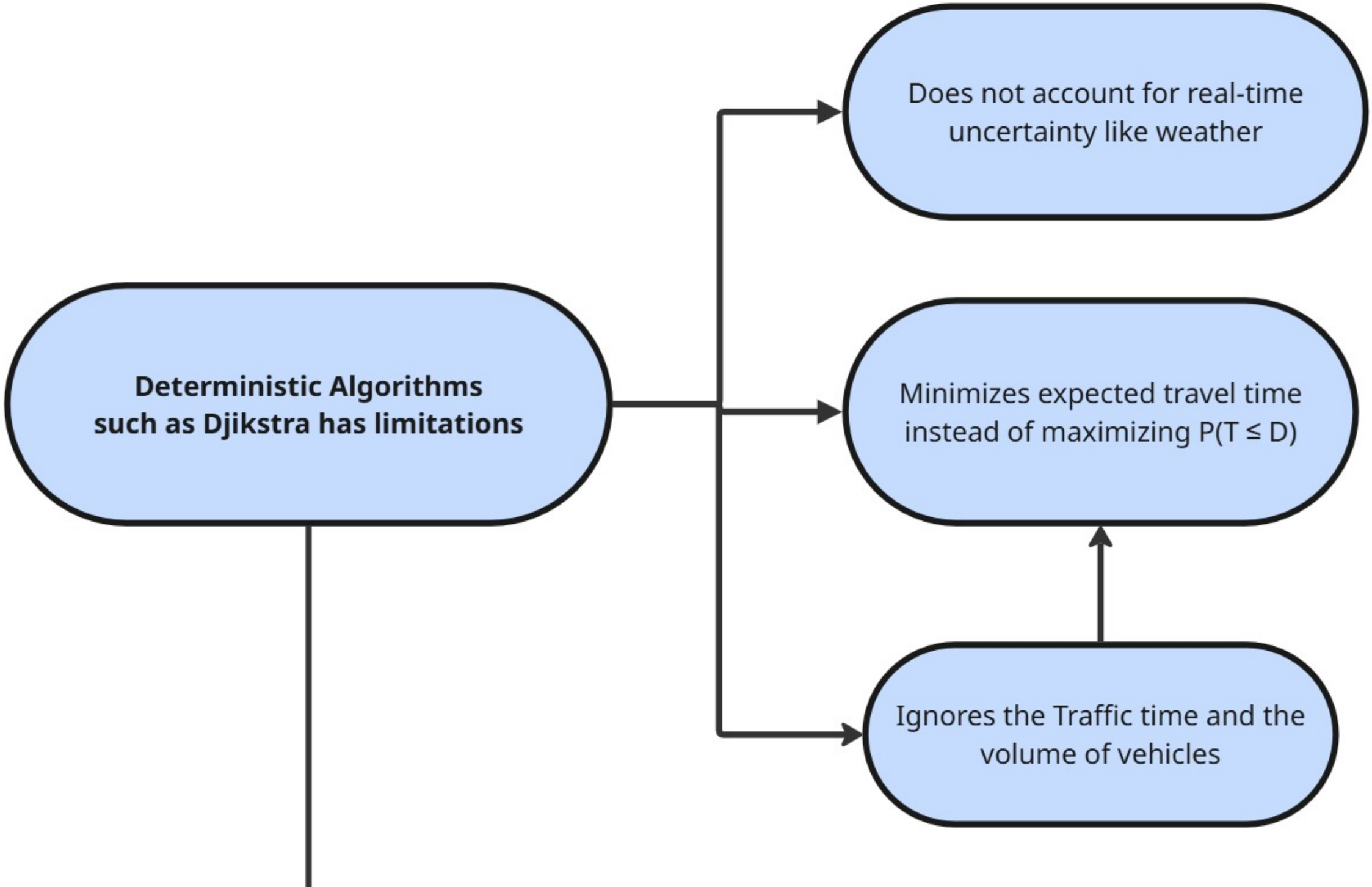
Milestone-03 Specific

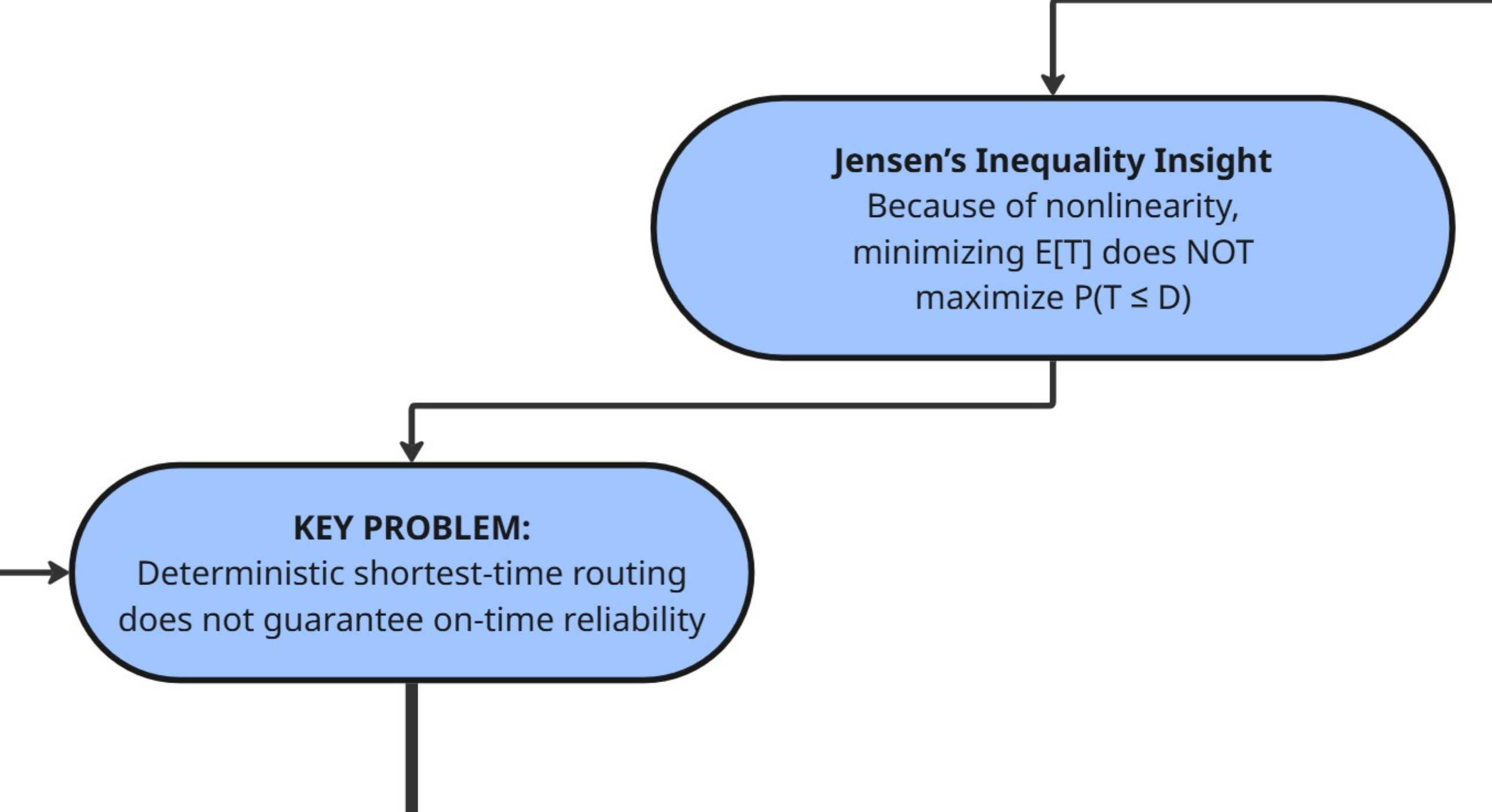


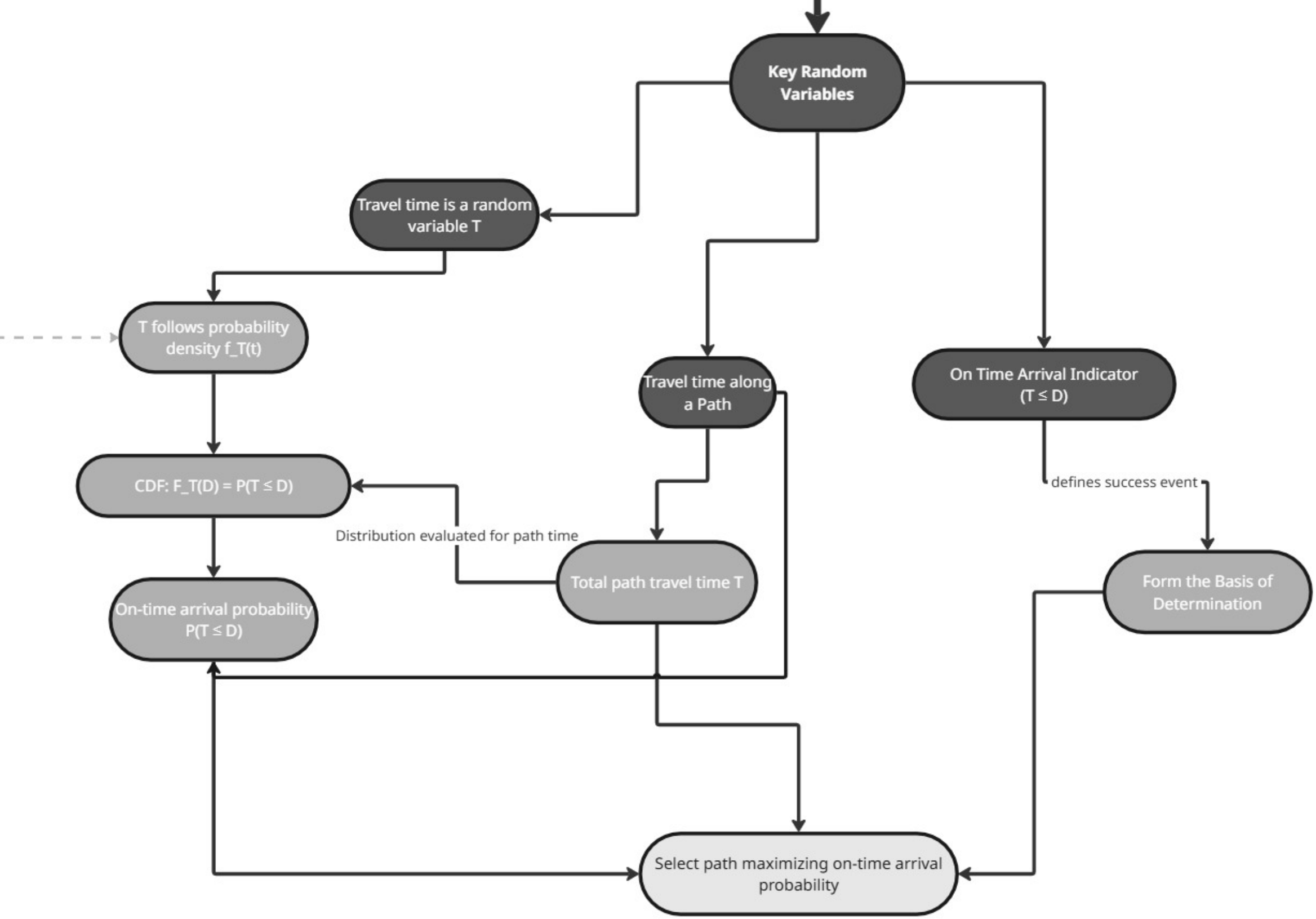

Main Objectives
Instead of minimizing expected travel time, focus on reaching the destination within time **D**
Start from destination and go to source by comparing the **Remaining Time Budget**
Determine the entire path before departure
Maximize the Probability of reaching the destination within the **time budget D**

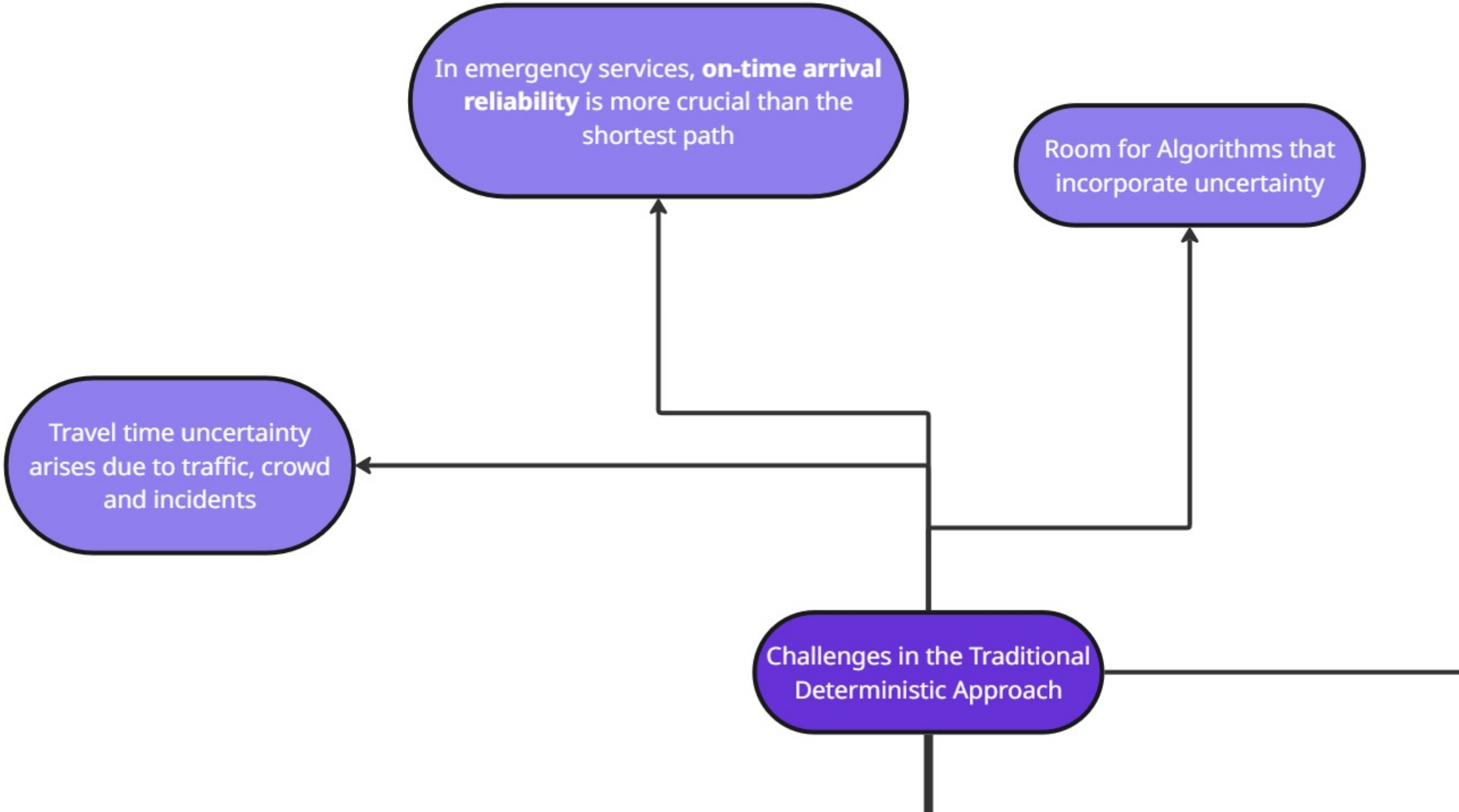

Sources of Uncertainty
Random travel time on each road segment
Traffic congestions and envrionmental uncertainty
Historical or Modeled data is used for Time Distributions


Models edge travel time as random variable T
Uses PDF/CDF of travel time
Our Approach: Path based SOTA (Stochastic On Time Arrival)
Selects most reliable path (not necessarily shortest)
Outputs most reliable path maximizing P(T ≤ D)
Road network modeled as a directed graph


Deterministic Algorithms such as Djikstra has limitations
Does not account for real-time uncertainty like weather
Minimizes expected travel time instead of maximizing P(T ≤ D)
Ignores the Traffic time and the volume of vehicles


Jensen's Inequality Insight
Because of nonlinearity, minimizing E[T] does NOT maximize P(T ≤ D)
KEY PROBLEM:
Deterministic shortest-time routing does not guarantee on-time reliability


Key Random Variables
Travel time is a random variable T
T follows probability density f_T(t)
Travel time along a Path
On Time Arrival Indicator (T ≤ D)
CDF: F_T(D) = P(T ≤ D)
defines success event
Distribution evaluated for path time
Total path travel time T
Form the Basis of Determination
On-time arrival probability P(T ≤ D)
Select path maximizing on-time arrival probability


In emergency services, **on-time arrival reliability** is more crucial than the shortest path
Room for Algorithms that incorporate uncertainty
Travel time uncertainty arises due to traffic, crowd and incidents
Challenges in the Traditional Deterministic Approach

Conclusion:
Overall better in complexity and comparison power but not feasible for logical and real life decisions
Least Expected Time
Always choose the shortest path based on the PMF and relative PDF
A subset or variance of Serious Dijkstra Algorithm
Comparably less logical and requires less time complexity
More faster than SOTA approach (A* Heuristic)
LET Approach Milestone-03
Specific
In emergency services, on-time arrival reliability is more crucial than the shortest path
Room for Algorithms that incorporate uncertainty
Travel time uncertainty arises due to traffic, crowd and incidents
Challenges in the Traditional Deterministic Approach
Deterministic Algorithms such as Dijkstra has limitations
Does not account for real time uncertainty like weather
Minimizes expected travel time instead of maximizing P(T ≤ D)
Ignores the Traffic time and the volume of vehicles
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Because of nonlinearity, minimizing E[T] does NOT maximize P(T ≤ D)
KEY PROBLEM:
Deterministic shortest-time routing does not guarantee on-time reliability
Instead of minimizing expected travel time, focus on reaching the destination within time D
Main Objectives
Start from destination and go to source by computing the Remaining Time Budget
Determine the entire path before departure
Maximize the Probability of reaching the destination within the time budget D
Stochastic On Time Arrival (SOTA)
Maximize P(T <= D) under uncertainty
Our Approach Path based SOTA (Stochastic On Time Arrival)
Sources of Uncertainty
Random travel time on each road segment
Traffic congestion and environmental uncertainty
Historical or Modeled data is used for Time Distributions
parameter estimation
Key Random Variables
Travel time is a random variable T
T follows probability density f_T(t)
CDF: F_T(t) = P(T ≤ t)
On-time arrival probability P(T ≤ D)
Travel time along a Path
Total path travel time T
Distribution evaluated for path data
On Time Arrival Indicator (T ≤ D)
defines success/failure
Form the Basis of Determination
Select path maximizing on time arrival probability